Public Court Data Pipeline - M2 Progress Report

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Source	Records	Endpoint	Status
CourtListener Rest API	600	/api/rest/v4/clusters/	Operational
CAP Georiga Bulk (Vol. 1)	93	data/raw/cap/json/	Operational
Total	693	Parquet + CSV – SQLite	Verified

What works

- **CourtListener ingestion** - 600 federal court records retrieved across scotus, ca1, and ca9 via the clusters REST endpoint with retry logic and pagination.
- **CAP bulk ingestion** - 93 Georgia state court records loaded from Volume 1 zip archive (Harvard Caselaw Access Project), parsed from individual JSON files.
- **Schema normalization** - Both sources have been mapped to a combined 10 column schema (case_id, court, jurisdiction, filing_date, decision_date, case_name, document_count, source, year, month) enforced via pandas DataFrames in db_handler.py.
- **Persistent storage** - All 693 records written to partitioned Parquet (by source/year/month), flat CSV, and SQLite. Data stays between runs and reads back cleanly through a verify mode.
- **Pipeline orchestration** - main.py supports five modes: test, cl, cap, full, and verify. Single entry point can run any stage.
- **Configuration management** - No hardcoded paths or credentials. All settings flow from config, settings and env.

Challenges Encountered

- **CourtListener endpoint mismatch:** The M1 plan targeted the opinions endpoint, but the v4 API returns richer cluster metadata via /clusters/. Resolved by switching endpoints and updating pagination logic.
- **CAP date format variability:** Georgia records use partial dates (1846-03) without a day component, causing year/month extraction to fail. Resolved with a null fallback that stores year=0, month=0 for non parseable dates.

- **Parquet categorical type bug:** Pandas categorical columns caused a min/max computation error in the normalizer. Resolved by casting year to numeric before aggregation.
- **Parquet append behavior** Re-running the pipeline appended new partitions rather than overwriting, inflating row counts across runs. Resolved by setting existing_data_behavior='overwrite_or_ignore' in pyarrow.

Changes from M1

- Endpoint pivot Switched from /opinions/ to /clusters/ on CourtListener for richer metadata coverage.
- Court scope reduced Originally planned four courts (scotus, ca1, ca9, dcd). Dropped dcd to stay within API rate limits; three courts fully operational.
- CAP scope reduced M1 proposed full Georgia bulk ingestion. M2 loads Volume 1 only (93 records) due to local storage constraints. Additional volumes can be added by placing more zip files in data/raw/cap/.
- Spark deferred to M3 M1 architecture included PySpark and HDFS for distributed processing. These remain planned for M3; M2 uses pandas for local normalization with a cloud-ready partitioned Parquet output format.

Next Steps for M3

- PySpark integration Replace pandas normalization with Spark DataFrames for distributed processing.
- Spark SQL analytics Distributed query execution over partitioned Parquet for case volume trends, court activity summaries, and jurisdiction comparisons.
- MapReduce aggregations Map/Shuffle/Reduce over (court, jurisdiction, year) groupings to compute completeness statistics and metadata coverage rates.
- Expand CAP coverage Ingest additional Georgia volumes to increase state court record count.
- Update architecture diagrams Reflect M2 implemented components vs. M3 planned components.

Timeline Assessment

Status: On Track. All M2 deliverables are complete. Both data sources are fully operational, 693 records persist across three storage formats, the pipeline is modular and reproducible, and the repository is clean with no exposed credentials.

Milestone	Planned	Status
Api Authentication	M2	Complete
CourtListener Ingestion	M2	Complete
CAP Bulk Ingestion	M2	Complete
Schema Normalization	M2	Complete
Data stored in Parquet, CSV, SQLite Storage	M2	Complete
Environment/Dependencies Documented	M2	Complete
Swap pandas to PySpark DataFrames	M3	Planned
MapReduce aggregations by court, jurisdiction, year	M3	Planned
Case volume trends over time	M3	Planned
Metadata completeness statistics per field	M3	Planned
Expand CAP to more Georgia volumes	M3	Planned